Automated Parallel Raman Microspectroscopy using Programmable Multipoint Illumination

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Abstract

Raman spectrum provides information about molecular content of a sample on the basis of inelastic scattering of light. However, since only one in ten million photons undergo inelastic scattering, measuring Raman spectra from multiple points is a time consuming process. In this paper, we present an automated parallel Raman microscpectroscopy scheme for simultaneous collection of Raman spectra from multiple points. Interest points from the samples were automatically detected using image processing algorithms and multiple point illumination patterns were generated automatically based on system-defined constraints. Also, hologram generation for multiple points was parallelized using Graphics Processing Unit (GPU). The illumination patterns guarantee complete sample coverage based on user defined parameters. We demonstrate that the total time taken for the complete process is in the order of a thousand milliseconds and is independent of the number of points to be simultaneously illuminated and thereby the density of the sample.

***Index Terms*— Interest point detection, Parallel Raman, Hologram generation using Graphics Processing Unit.**

I. Introduction

Raman spectroscopy is used in many fields dedicated towards non destructive analysis of materials [1] for studying their molecular composition and structure. Raman spectrum is collected from inelastically scattered photons. However, since such scattering is very rare (one in ten million), Raman imaging has very low time resolution [2]. Traditionally, Raman spectra from multiple points in a field are collected on a point-by-point basis (known as point-scan method [3]). Since the Raman spectra is a weak phenomena the laser dwelling time at each point may vary from millisecond to seconds and in addition data needs to be read out after each point acquisition which typically adds another few hundred seconds. Depending on imaging parameters and the number of points of interest, this may take up to a few hours. An improvement over this is a line-scan method. In this method, a single laser beam is used to create multifoci excitations and multiple spectra are collected simultaneously. Image acquisition is ~40 times faster than in point-by-point scan [3]. Further reduction in imaging time could be achieved by using a parallel acquisition scheme [4] based on either time-sharing or power sharing mechanism. In time sharing method, laser is rapidly scanned over multiple points during the time of single CCD recording frame. In power sharing method, laser is shaped into an elongated line and the entire line is imaged by single CCD recording

frame. The temporal power fluctuation of each method within a CCD frame is different. When the points are not in a line time sharing methods provide more flexibility and are particularly advantageous for sparse samples like bacteria.

In this paper we present a fully-automated parallel Raman microspectroscopy scheme which is an improvement over the power-sharing approach proposed in [4]. Specifically, we designed and integrated robust image processing algorithms for automated interest-point detection and automated illumination pattern generation. We also use a GPU-optimized hologram generation algorithm which scales down the imaging time by an order of magnitude.

Experimental setup shown in figure 1 is a power sharing approach described by [4] which allows simultaneous imaging of multiple points not aligned in a horizontal line. Programmable multipoint illumination with wide field Raman imaging is used. The setup consists of 785 nm output of continuous wave Titanium: Sapphire Laser which in turn is filtered by Laser line-filter (SemrockLL01-785-12.5) and expanded to ~1cm diameter before the spatial light modulator ((SLM) liquid crystal on silicon (LCOS) Hamamatsu). An inverted microscope (OlympusIX71) is used and the output from the SLM is fed through the back port with an addition of tube lens. A dichroic mirror (Semrock LPD01-785RU-25) is placed in the microscope turret for epi-Raman acquisition via a microscope objective (Olympus UPLSAPO 60XW, 1.2 NA). The main disadvantages of this method are the interest points detected are fed manually through an interface in Lab View. The illumination patterns are generated manually which is very tedious when the sample is not sparse. Also, hologram generation is not GPU-optimized which tends to become very slow as the number of interest-points increases.

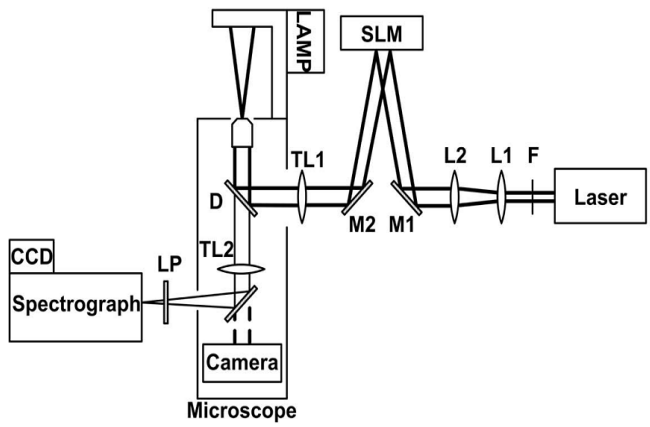

Figure 1: Experimental setup: laser line filter (F); beam expander (L1/L2); spatial light modulator (SLM); mirrors (M1/M2); tube lenses (TL1/TL2); dichroic beam splitter (D); long-wave pass filter (LP).

II. METHODS

In order to make the Raman spectrum collection fully automated, we modified the setup as shown in figure 2. Specifically, the modified setup has an additional unit which contains an Image Processing Module, Automated Pattern Generation Module and Hologram Generation module which are described in the remainder of this section.

Image Processing Module

A detailed view of this module showing all image processing filters employed is shown in figure 3. We refer to this as the image processing pipeline.

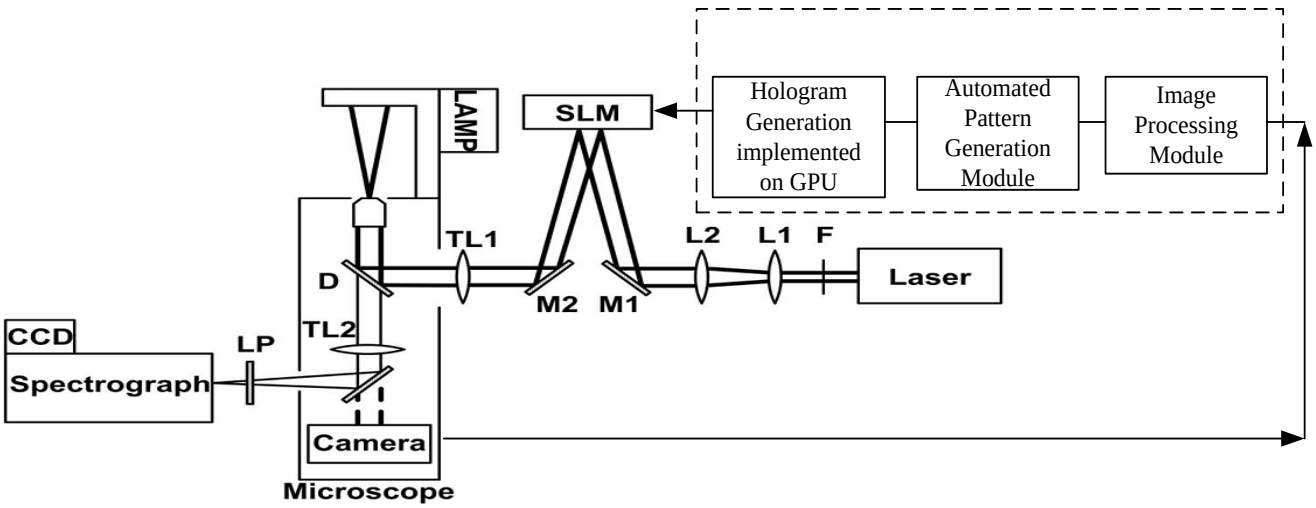

Figure 2: Modified experimental setup showing Image Processing Unit; Automated Pattern Generation Module; and Gerchenberg-Saxton algorithm.

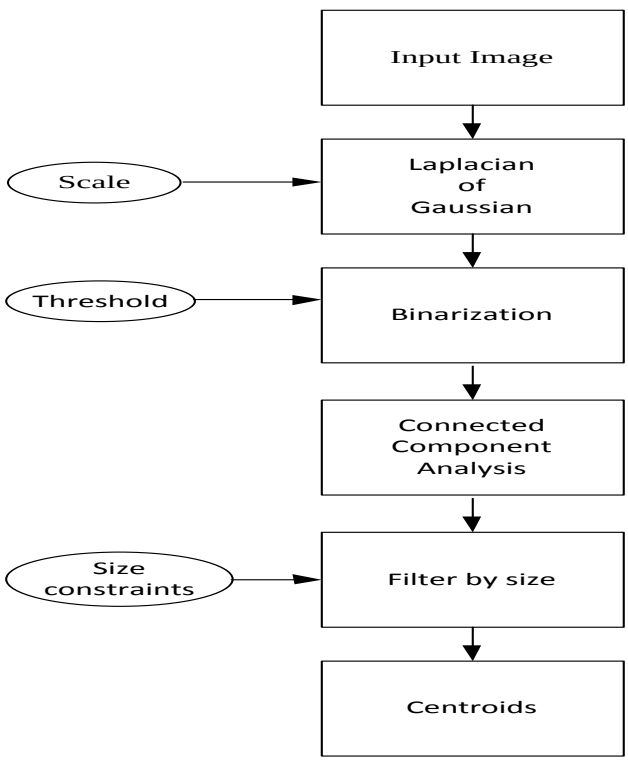

Figure.3: Image Processing module pipeline.

The input full-field image (example, figure 4) is passed to the first block of the pipeline which uses the Laplacian of Gaussian (LoG) filter. The LoG kernel tries to identify points in the image that differ in properties like brightness or color from the surrounding [5 cite]. The LoG of an image is the convolution of Laplacian and Gaussian kernels with the image. Gaussian kernel is used for smoothing the image and Lalplacian is to extract points of interest. Laplacian of Gaussian kernel is formulated as,

$$LoG = \Delta G_\sigma(x,y) = \frac{\partial^2}{\partial x^2} G_\sigma(x,y) + \frac{\partial^2}{\partial y^2} G_\sigma(x,y) = \frac{x^2+y^2-2\sigma^2}{\sigma^4} e^{\frac{-(x^2+y^2)}{2\sigma^2}}$$

where Sigma 'σ' represents the scale at which the LoG operator finds the points of interest. This scale is correlated to the scale of the objects in the sample and can be set as a user defined parameter in our system. For our experiments, scale of three units gave us decent results. The output of the LoG filter is shown in figure 5 (left). Note that the LoG filter also gives high response at the shadows of the objects. The next stage of the pipeline is conversion of the gray scale image to binary image. Since the LoG filter gives a robust response around interest points, this is not a critical parameter and is not

exposed to the user. The result of this step is shown in figure 5 (right). This is followed by labeling every object in the image by a unique number (connected component analysis) as. The labeled objects are relabeled based on size for filtering out very small and very large objects which might be false positives from the earlier steps. Figure 6 shows the result after filtering the relabeled image. Every object now has a color corresponding to the size of the object. Finally the centriods of the objects are calculated (figure 7 shows a magnified image with centroids shown in pink). These centroids are our interest points, the points where laser should be focused.

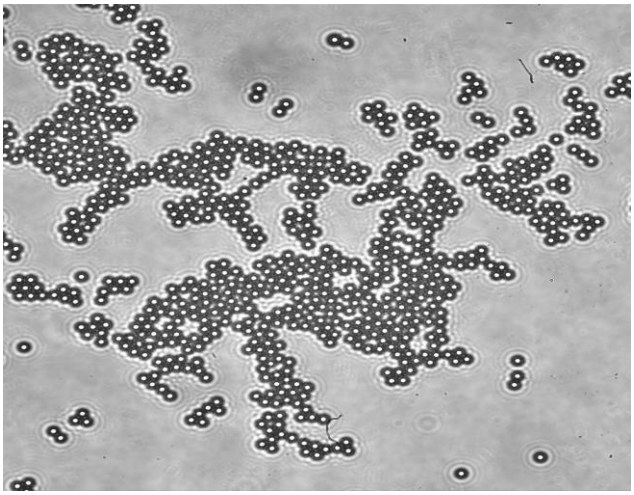

Figure 4: A sample input image, 2 micrometer Polystyrene beads.

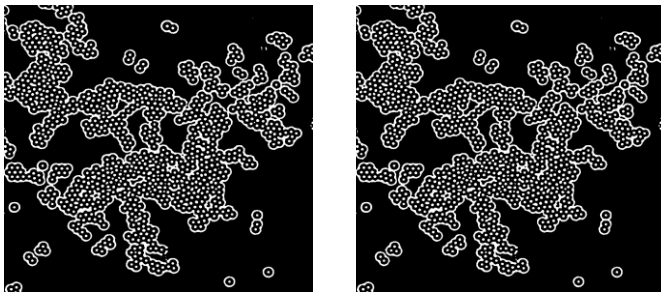

Figure 5: Output of LoG filter (left) and binarization filter (right).

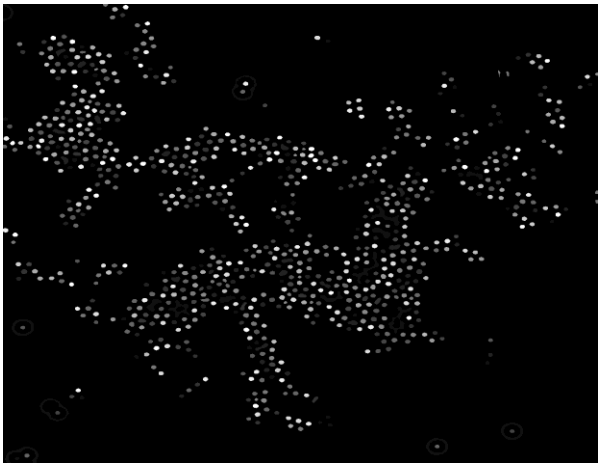

Figure 6: Output after size filtering: objects are labeled and filtered by size.

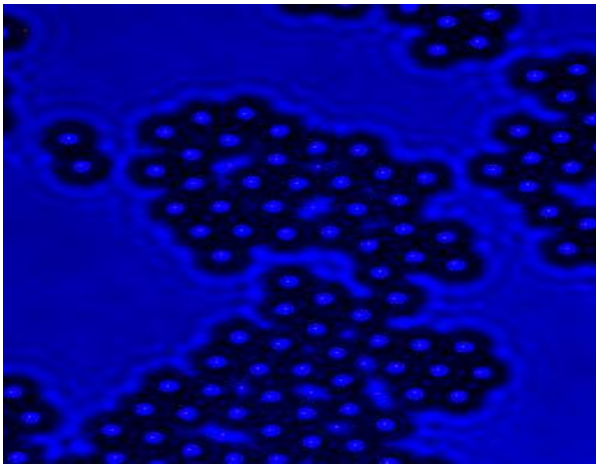

Figure 7: Image (magnified) showing the computed centroids in pink.

Automated Pattern Generation Module

To allow easy unmixing of spectra collected from multiple points, the current grating settings required us to not have interest points aligned horizontally. This requirement is satisfied by automatically generating illumination patterns. This algorithm finds the maximum number of points that can accommodate within one pattern in the ascending order of the size of the objects. This is repeated until all the points are covered. Figures 8 shows a few sample patterns generated by this algorithm.

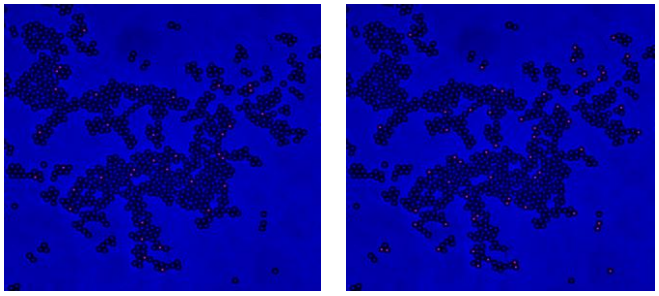

Figure8: Example sample patterns generated according to system-defined constraints.

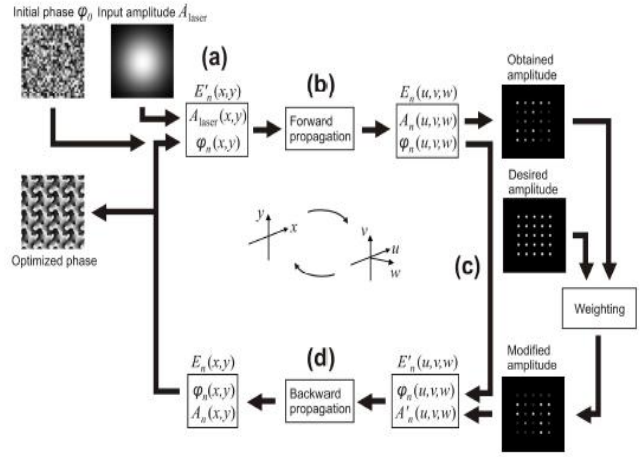

Figure 9: Block diagram for Gerchenberg-Saxton algorithm.

Hologram Generation Module

For computing the holograms, we used code developed by Martin Presson's group [6]. In this code, the hologram generation is based on Gerchenberg-Saxton (GS) algorithm (figure 9) and is implemented on a GPU using CUDA (Unit of 448 cores and 144GB/s bandwidth, NVIDIA® Quadro® 6000). For a particular set of spots, the algorithm generates their corresponding hologram, that is, the phase pattern required by the SLM to focus light at the given spots [7]. Initially a random guess of phase distribution is used and complex field in the spot positions is computed by Fourier transform. If the amplitude of the obtained field doesn't correspond to certain specified requirements the amplitude is replaced using certain weighting function and this is repeated until an optimized phase which in turn gives desired amplitude is obtained. Number of iterations and other such important parameters for this code are exposed to the user. Figure 10 shows the holograms generated by this code for different number of points. These holograms are passed to the SLM to obtain the desired light field distribution.

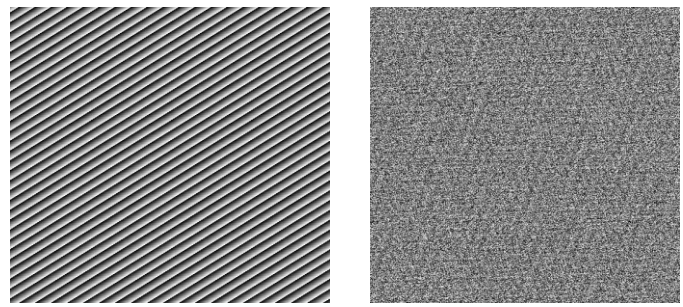

Figure 10: Sample holograms generated by the GS algorithm for different number of input points.

III. Results and Discussion

We report the time spent in generating the hologram only as a function of number of spots (table 1). The graphic for this is shown in figure 11. Again, the time taken is almost constant as a function of number of points with a mean of about 6 microseconds. This demonstrates the acceleration provided by GPU. Other static time related to CUDA are shown in table 2. The total computational time required by all our modules is about 1.2 seconds. We also obtained similar results on spore images (not shown here). If the rate at which a hologram is generated can be maintained to

be low, (depending on the application) tracking can be easily made possible. One quick way of achieving some more time savings is by implementing the image processing filters on the GPU using CUDA.

Table1: Time analysis of the code with and without disk IO.

Number of spots	**Time taken to generate pattern and Hologram (sec)**	**Time taken to generate pattern and Hologram (including IO) (sec)**
0	0	0.145
100	0.007	0.143
200	0.007	0.152
300	0.006	0.144
400	0.007	0.142
500	0.008	0.140
600	0.007	0.152
700	0.008	0.144
800	0.008	0.143
870	0.006	0.143

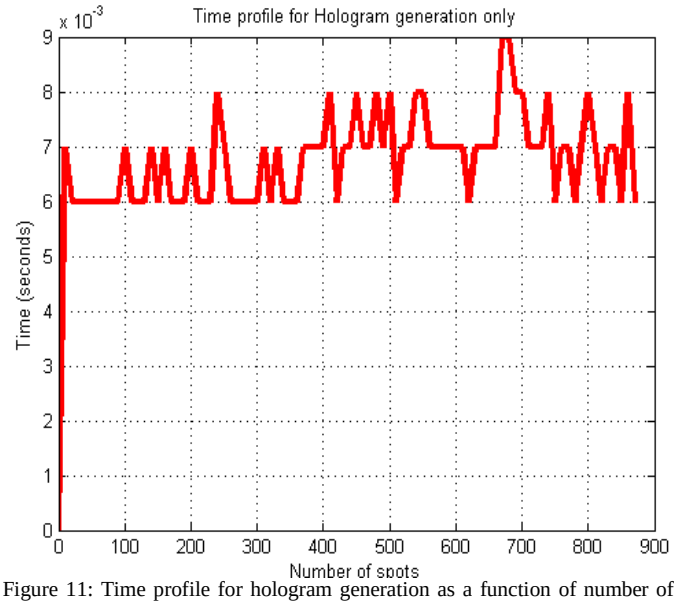

Figure 11: Time profile for hologram generation as a function of number of spots.

Table2: Time analysis of the code for CUDA related static functions.

Module	**Time (sec)**
Load Library	0.007 sec
Start 'GenerateHologramCuda.dll'	0.173 sec
Stop 'GenerateHologramCuda.dll'	0.017 sec
Free Library	0.002 sec

IV. Conclusion And Future Work

We conclude that our method is fully automated and optimized in terms of time. The algorithm consumes very less time and works well for dense samples. Time for hologram generation remains almost same as the number of spots increases. Applications of this technique for measurement of Raman spectra for biological samples like bacteria should be straight forward. Applications for collecting Raman spectra as a function of time for live cells can also be explored.

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References

[1] John R. Ferraro, Kazuo Nakamoto and Chris W. Brown, "Introductory Raman Spectroscopy", ISBN: 978-0-12-254105-6 (2003).

[2] Lingbo Kong, Pengfei Zhang, Jing Yu, Peter Setlow, and Yong-qing Li, "Rapid confocal Raman imaging using a synchro multifoci-scan scheme for dynamic monitoring of single living cells" Appl. Phys. Lett. 98, 213703 (2011).

[3] Ji Qi and Wei-Chuan Shih, "Parallel Raman microspectroscopy using programmable multipoint illumination," Optics Letters, Vol. 37, Issue 8, pp. 1289-1291 (2012).

[4] Rafael C. Gonzalez, Richard E. Woods, "Digital Image Processing", 2nd Edition, ISBN-13: 978-0201180756 (2003).

[5] https://github.com/MartinPersson/HOTlab

[6] Martin Persson, David Engström, Anders Frank, Jan Backsten, Jörgen Bengtsson, and Mattias Goksor, "Minimizing intensity fluctuations in dynamic holographic optical tweezers by restricted phase change" Optics Express, Vol. 18, Issue 11, pp. 11250-11263 (2010).